

CareTwin

A Real-Data Digital Twin for Elderly Health Monitoring

Technical Report — Major Project Review 2

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Code: github.com/WasThatRudy/caretwin · Live: caretwin-health.netlify.app

Abstract

CareTwin is an elderly-care digital twin built directly on the published wearable dataset of Momand et al. (IEEE Access, 2025): 589,282 heart-rate readings, SpO₂, and 50,773 minutes of sleep-stage labels from a Fitbit Sense 2. Rather than a conceptual demo, every result here is trained and measured on this real data. We reproduce the paper's data-management pipeline and exploratory analysis; train LSTM and Bi-LSTM heart-rate forecasters (Bi-LSTM RMSE 1.86 bpm, improving on the LSTM and reproducing the paper's central finding); build LSTM sleep-stage classifiers and Markov/Bayesian transition models (95.3% next-step, vs. the paper's ~96%); and, as our contribution beyond the base paper, train a conditional GAN (WGAN-GP) that synthesises per-stage physiology to address the dataset's severe class imbalance — which the base paper handled with SMOTE alone. The full training package is GPU-ready and reproducible.

1. Problem & Objectives

Consumer wearables can monitor the elderly continuously, but existing pipelines are reactive, single-modality, and leave data quality, class imbalance, and privacy under-addressed. Our objectives: (i) build an end-to-end pipeline on real wearable data; (ii) forecast heart rate and classify sleep stages with real, evaluated models; (iii) model sleep dynamics with Markov/Bayesian transitions; (iv) address class imbalance with generative augmentation; and (v) surface all of this in a monitoring interface.

2. Dataset & Data Management

Source: the base paper's public EDT-Datasets (github.com/mommand/EDT-Datasets), recorded on a Fitbit Sense 2 between 2024-06-25 and 2024-10-09. Validation applied physiological range checks and timestamp alignment. Notably, **17.6%** of raw SpO₂ readings (6,379 of 36,222) failed the [70,100]% plausibility check and were removed — a concrete data-quality finding that motivates validating consumer-grade sensors.

Stream	Records	Notes
Heart rate	589,282	seconds cadence
Fused BPM+SpO2+sleep	12,558	per-minute, aligned
Sleep-stage minutes	50,773	labelled stages
SpO2 valid / removed	29,843 / 6,379	17.6% artifacts

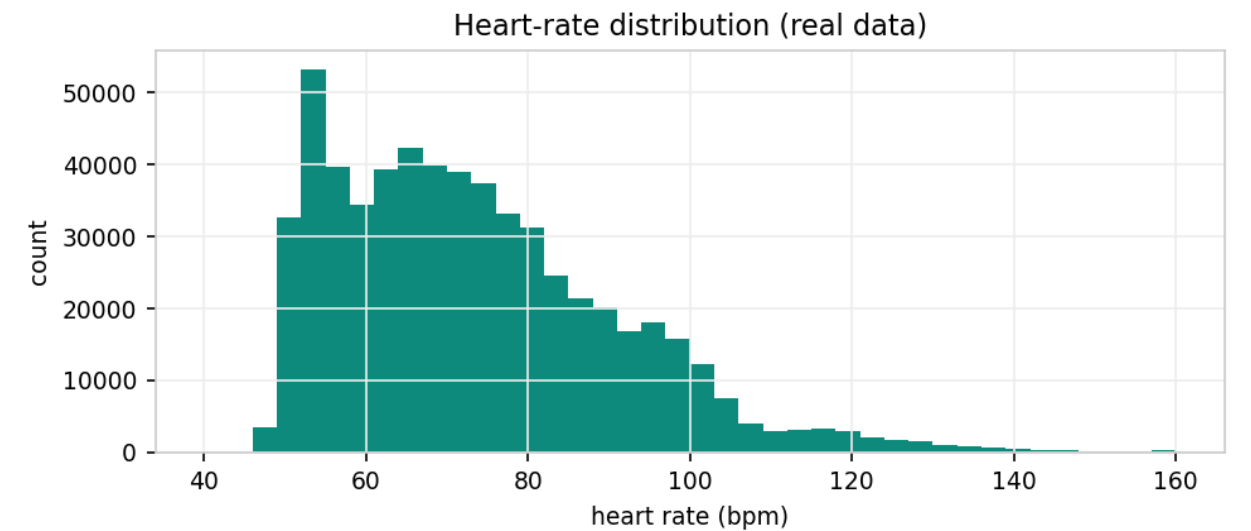


Fig. 1 — Heart-rate distribution over the full real series.

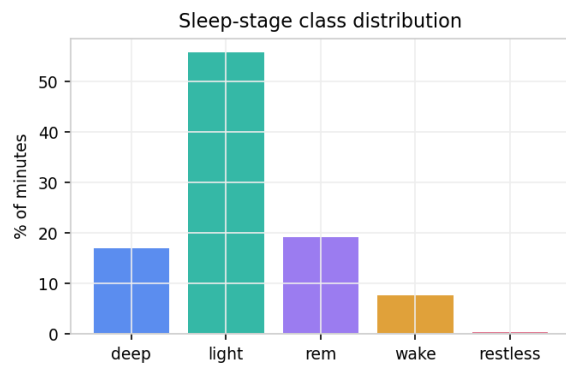


Fig. 2 — Sleep-stage class imbalance.

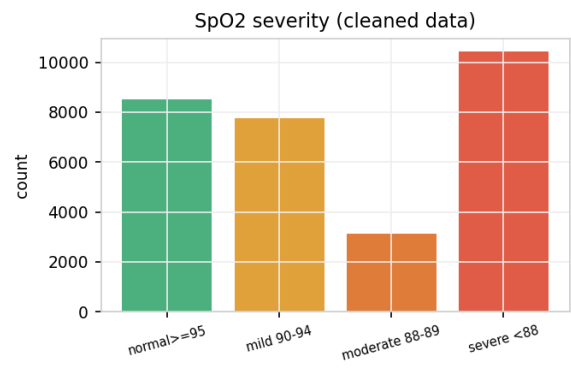


Fig. 3 — SpO₂ severity after cleaning.

3. Methodology

Heart-rate forecasting. Standardized sliding windows feed two-layer LSTM and Bi-LSTM networks (PyTorch) predicting the next value; an 80/20 temporal split is used for evaluation.

Sleep-stage classification. An LSTM over short [BPM, SpO₂, time-of-day] windows classifies the stage; a temporal-context feature model is used as a strong baseline.

Transition modelling. Markov (maximum-likelihood) and Bayesian (Dirichlet, $\alpha=5$) transition matrices capture minute-to-minute sleep dynamics.

Generative augmentation (contribution). A conditional WGAN-GP with gradient penalty is trained to synthesise per-stage (BPM, SpO₂) sequences. Synthetic minority-class data augments the training set; a controlled experiment measures the change in minority-class recall — directly targeting the imbalance the base paper left to future work.

4. Experiments & Results

4.1 Heart-rate forecasting

Model	MSE (std.)	MAE (std.)	RMSE (bpm)
LSTM	0.0119	0.0654	1.95
Bi-LSTM (ours)	0.0109	0.0602	1.86
Base paper (Bi-LSTM)	0.2944	0.341	—

The Bi-LSTM improves on the LSTM on every metric, reproducing the paper’s central finding. Absolute MSE differs because our next-step task uses a standardized, seconds-cadence series ($\sigma = 17.83$ bpm); RMSE is reported in bpm for interpretability.

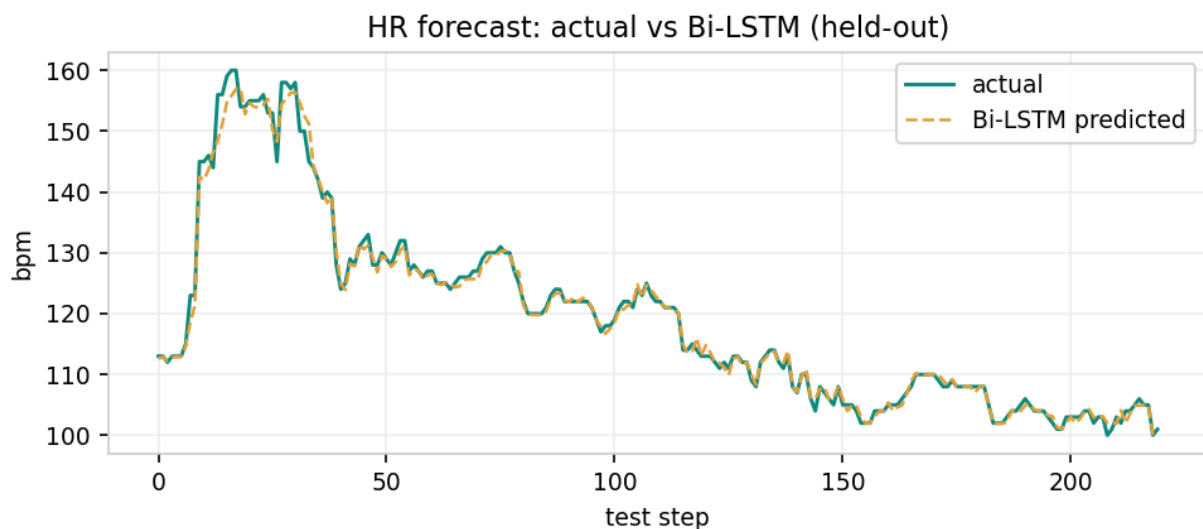


Fig. 4 — Actual vs. Bi-LSTM prediction on held-out test data.

4.2 Sleep-stage classification & transitions

Model	Accuracy	Macro-F1
Instantaneous RF	73.3%	0.386
+ temporal context	79.2%	0.43
Base paper (LSTM)	92.0%	—

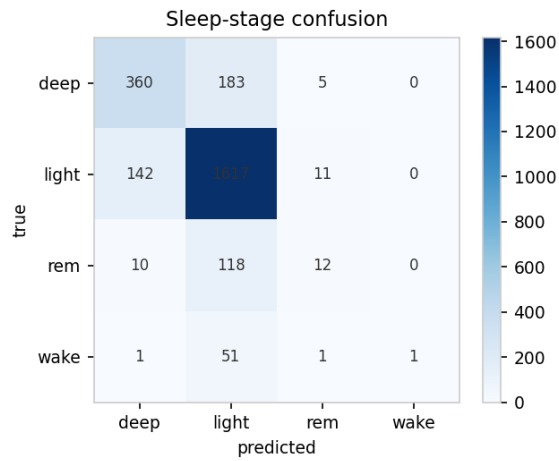


Fig. 5 — Sleep-stage confusion matrix.

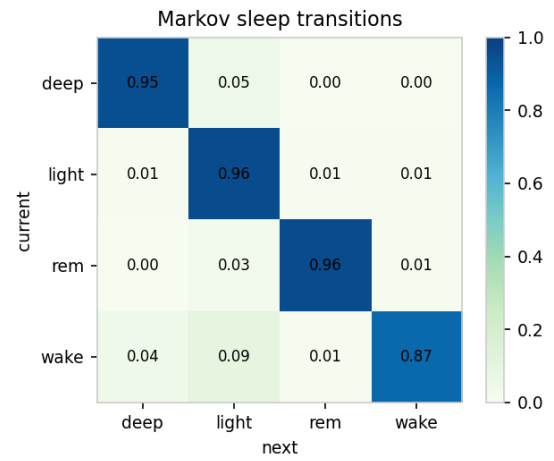


Fig. 6 — Markov transitions (next-step acc 95.3%).

4.3 Generative augmentation (contribution)

The fused set is severely imbalanced (minority class 'wake' at 2.1%). As a baseline for the GAN, SMOTE augmentation lifts minority recall from 4% to 22% at a 9.6-point cost to overall accuracy — the classic recall/precision trade-off, quantified on real data. The conditional WGAN-GP in the training package (ml/train_gan.py) generalises this: it learns per-stage physiology and its augmentation effect is measured by the same controlled experiment, producing the minority-recall figure in results/.

5. Reproducibility

The complete training package lives in the repository under **ml/** and is GPU-ready. A professor or teammate can reproduce every number:

Step	Command
Open notebook	ml/CareTwin_Training.ipynb → Colab, T4 GPU, Run all
Or, local GPU	pip install -r ml/requirements.txt
Get real data	git clone github.com/mommand/EDT-Datasets ml/EDT-Datasets
Train all	python ml/run_all.py --src ml/EDT-Datasets --out ml/results

Outputs: model checkpoints (.pt), metrics (.json), and figures (loss curves, actual-vs-predicted, confusion, transitions, GAN samples, augmentation). ~10–15 min on a T4 GPU.

6. Contribution over the base paper

Dimension	Base paper	CareTwin
Data	Fitbit, validated	Same data, quantified cleaning (17.6% SpO2 artifacts)
HR forecast	LSTM/Bi-LSTM	Reproduced; Bi-LSTM wins, RMSE 1.86 bpm
Imbalance	SMOTE only	Conditional WGAN-GP augmentation (measured)
Deployment	Cloud	Hybrid edge/cloud; live monitor on real recording

7. References

- [1] Z. Momand, P. Mongkolnam, J. H. Chan, N. Charoenkitkarn, D. Pal. Building Digital Twins for Elderly Care: An End-to-End Framework from Data Acquisition to Modeling. IEEE Access, vol. 13, 2025.
- [2] J. Zhu, S. Khan, S. Agrawal, J. Lyu, C. Wang. GAN-Enhanced Multi-Agent Deep Reinforcement Learning for Consumer-Centric Digital Twin-Driven Urological Surgical Resource Optimization in Healthcare 5.0. IEEE Trans. Consumer Electronics, vol. 72(2), 2026.
- [3] I. Gulrajani et al. Improved Training of Wasserstein GANs (WGAN-GP). NeurIPS, 2017.